

$$\text{NLL} = -\log \left(\sum_i \exp((\mathbf{l} + \mathbf{m})_i) \right) \quad (7)$$

Where: $\mathbf{l}$ is the log likelihood of the circuit, $\mathbf{m}$ is the mixing value decided by the number of repetitions.

4.4 Methodology

4.4.1 Considerations for combining NeSy with HTC

The implementation of NeSy models like SPL in text classification tasks may require some modifications, including the base model, the circuit logic, and other hyperparameters.

Base model

Traditionally, NeSy architectures are built upon neural networks such as ResNet, a popular choice for computer vision applications. SPL, however, was tested on much simpler tasks, hence a minimalist neural network was used as the base model. Understandably, this architecture is not expressive enough to process BERT embeddings (informal tests yielded a 10% accuracy at best). This implies that NeSy methods need to be adapted to an LLM base for an equal footing with their competitors. In this study, the base model is left unfrozen. This allows the model to finetune similarly to the baseline.

Circuit modifications

Let the constraint circuit of SPL be α . α is made by conjoining all individual constraints (β) for each node. The original logical formulas for SPL classification tasks are as follows:

$$\alpha = \beta_1 \wedge \beta_2 \wedge \dots \wedge \beta_n \quad (8)$$

$$\beta_i = \neg i \vee (((\beta_1 \wedge J_1) \wedge J_2) \wedge \dots \wedge \beta_n) \quad (9)$$

Where: i and j are ordered nodes in the hierarchy, and j is a child of i .

In informal tests, the original circuit of SPL was not sufficient in constraining power for these text classification tasks, as it still showed numerous errors related to mutual exclusivity (ME) - where over or under one class is selected for each level, therefore

possibly explaining the poor initial results.

The logical formula concerning β has been modified as:

$$\beta_i = \neg \vee (((\beta_1 \vee J_1) \vee J_2) \vee \dots \vee \beta_n) \quad (10)$$

Here, β serves to maintain parent-child relationships: β entails at least one of its children, instead of all of them.

Completeness of hierarchies

Next, the completeness of each hierarchy also poses significant considerations in labelling, preprocessing, and circuitry.

A hierarchy is considered complete when every sample is annotated with exactly one class on each level. An incomplete hierarchy, however, may have missing annotations or duplicate answers (over one annotation per level). The current study involves two datasets with complete hierarchies and one without.

We have taken the following steps to tackle this issue:

- All examples with duplicate annotations have been filtered from all splits. This leaves both training and testing free from violating samples, which will result in a difference in performance from the literature.
- For multihot-encoded labels, we have decided to include parent nodes as possible options for prediction. This in turn will mean that hierarchy violation checks will not be restricted to leaf nodes only, even for complete hierarchies (a missing leaf node may still count as no violation).
- The level to which constraints are applied will also be affected. We decided to weaken the non-zero (NZ) constraint for all datasets. Instead of applying to all levels (theoretically ideal for complete hierarchies), NZ now only applies to the root (we retained some samples with only root-level annotations).

The effect of leaving in parent nodes for prediction will be later discussed in Ablation Test 1 (Hard enforcement).

4.4.2 Computational graphs in BERT-MATCH and SPL

Graphs are a crucial component in many methods designed for hierarchical classification. They can serve a variety of purposes.

In the *htc-survey* implementation of BERT-MATCH, the hierarchy graph is used as a dynamic label for raw performance checks as well as for regularization.

First, the hierarchy is automatically learned by a multilabel binarizer based on the existing labels in the current split. It is then expressed as a directed graph (with distinction between parent and child).

The raw labels can be easily retrieved through inverse transformation. Hierarchical metrics such as AHC are then calculated by taking the mean of the distance traversed between true and predicted nodes on the graph.

This directed graph also allows BERT-MATCH to define the parent-child relationship for each type of regularization. It uses two regularization terms, namely output regularization, which rely on the difference in predicted probability (confidence) between the child and its parent, and parameter regularization, which is based on the difference between weight vectors assigned to these classes. They can be expressed in the form of formulas:

$$r_i = p_{i,c} - p_{i,p}, \quad i = 1, \dots, N \quad (10)$$

$$P_o = \lambda_1 \sum_{i=1}^N \max(0, r_i) \quad (11)$$

(1)

$$r_j = w_{p,j} - w_{c,j}, \quad j = 1, \dots, d \quad (12)$$

$$P_p = \frac{\lambda_2}{2} \|\mathbf{r}\|_2^2 \quad (13)$$

Where λ_1 and λ_2 are scaling factors.

This technique may pose a caveat, especially for incomplete hierarchies where annotations can be inconsistent: the multilabel binarizer may not be sufficient to maintain a complete hierarchy, since each split may not have the exact set of classes annotated, leading to uninformed predictions. Therefore, we propose another more rigorous method of binarizing: by extracting a multihot encoding external file which can be looked up. This

preprocessing step should allow for the binarization of all possible labels, hence more precise hierarchy graphs.

SPL also uses a directed graph, though instead of the binarizer, it is compiled with a taxonomy parser, which reads a taxonomy file and returns multihot encodings, as well as an ancestor matrix that encodes parent-child relationships. A layer mapping function is also used to get the level number of each node for more precise layer definitions. Aside from being used in the same performance metrics as BERT-MATCH, this graph plays a huge role in building the constraint circuit by providing the variables for logical operations. The layer map also helps define the specific layers where certain constraints are applied (see section below on SPL constraints).

4.4.3 The SPL-BERT constraint circuit

This circuit requires three logical constraints: β , δ and ζ to fully encapsulate the complexity of hierarchies, while adhering to the expectations of a meaningful prediction (as fully and clearly as possible). They shall now be detailed in order of appearance.

Zeta (ζ)

ζ is responsible for non-zero predictions, meaning that “this level must contain at least one prediction.” ζ can be expressed as a logical formula:

$$\zeta_i = \begin{cases} (\neg i \vee (\zeta_i \vee J_1) \vee J_2 \vee \dots \vee J_n), & \text{if } i \in \text{nz_layers} \\ 1, & \text{if } i \notin \text{nz_layers} \end{cases} \quad (14)$$

Where i is the level being processed, as defined by the layer map, and k (from 1 to n) is a node at level i .

Delta (δ)

δ manages over-one predictions, or in simpler terms, it ensures that “this level does not contain more than one prediction.”

The key logic behind δ lies in selecting one out of two variables for each variable combination of 2:

$$\delta \leftarrow \delta \wedge (\neg s_1 \vee \neg s_2) \quad (15)$$

where: L is the set of nodes at this level, and (s_1, s_2) are any combination of nodes in L

Together, ζ and δ enforce the model to only allow one prediction per level. These two constraints have to be conjoined before being integrated to the main graph in order to maintain both of their effects (if separately, they may cancel out one another). This pair of constraints help prevent mutual exclusivity errors by LLMs, which traditionally do not leverage symbolic systems for the validity of predictions.

Beta (β)

β is the constraint layer retained from the original implementation of SPL, though with significant changes. Here, it regulates parent-child relationships by enforcing that “if this node is predicted and it has a child, then at least one of its children must be predicted. If this node does not have a child, then this node is unrestricted by β .”

The β formula has been explained in the section above:

$$\beta_i = \neg i \vee (((\beta_1 \wedge J_1) \wedge J_2) \wedge \dots \wedge \beta_n) \quad (9)$$

In reality, the order of these constraints can be swapped: ζ with δ , and β with $(\zeta + \delta)$. Each constraint layer outrules a large amount of unsatisfying configurations. Theoretically, for a complete hierarchy, applying all three layers to every level should result in exactly n feasible configurations, where n is the number of classes (leaf nodes in the hierarchy).

For SPL-BERT, the BERT embeddings (concatenated from its three last layers, excluding the classifier) pass through the gating function, which learns the parameters to tune the circuit. The circuit then computes the Negative Log Likelihood (NLL) loss and outputs the Most Probable Explanation (MPE) based on the gates’ output. This MPE is used as the final output, while the loss is backwarded to both the base model and the gating function.

4.4.4 Datasets & Preprocessing

We considered all five datasets in the htc-survey for our experiments, but only three: Amazon 5x5 (AMZ), Blurb Genre Collection (BGC), and Web of Science (WOS) were implemented successfully. Due to time constraints, Reuters Corpus-V1 (RCV1) and Bugs were opted out of the current study. The details for their exclusion can be seen in Table

1.

	Bugs	RCV1-v2	WOS	BGC	Amazon
Size	35050	804414	46960	91894*	500000
Depth	2	4	2	4	2
Total labels	102	103	145	146	30
Split ratio (train:val:test)	53:13:34	3:0:97	59:12:29	64:16:20	53:13:34
MLP**	Yes	No	Yes	No	Yes
Reason for exclusion	Issues with data preprocessing	Time constraints due to requiring permission			

Table 1: Properties of the *htc-survey* datasets under consideration. Adapted from Zangari et al., 2024

* before filtering duplicate annotations

** Mandatory Leaf Prediction

Some of the datasets, such as AMZ or Bugs, had their labels sorted to be repurposed into HTC datasets. Of the 5 datasets, Bugs, BGC and RCV1 proved to be the bigger challenges for non-NeSy models, thanks to entirely different characteristics. Aside from the fact that both datasets require intensive preprocessing, Bugs offered highly specialized language alongside “noise” (grammatically incorrect sentences and low-level jargon), while BGC and RCV1 are challenging due to their depth and incompleteness of hierarchies.

Amazon 5x5

The AMZ dataset was created from scratch in the *htc-survey* paper using the 2018 Amazon dump by Ni et al., 2019. In total, 500000 product reviews (often the review text plus its title) were extracted and assigned multilabel annotations with a depth of 2. This dataset is fairly balanced, with exactly the same number of samples by first-level categorization. It is characterized by plain English (no particularly specialized classes) and clearly expressed sentiments, making it one of the easier datasets for LLMs like BERT.

Blurb Genre Collection

The BGC dataset proposed by Aly et al., 2019 was specifically tailored for HTC tasks. Each sample contains metadata, as well as a short description of book contents from the Penguin Random House publisher. Interestingly, it set out to mirror the RCV1 dataset in terms of structure, as seen in the non-mandatory leaf predictions and the same depth of 4. However, BGC is an unbalanced dataset and suffers from duplicate labels, turning

```

{
  'summary': '"Impressed for $10"',
  'reviewText': '"I picked up an old set of VW recaro bucket seats (the old gray ones). One of the cussions had a few holes in them (cigarette burn) so I decided I'd see what I could do. I worked on the seat in pieces so I did it inside. It could be done just as easily installed in the car. Simply use the color chart to mix a close match to the interior. Clean the area you will be working with. If necessary provide some backing such as a cotton ball stuffed down inside. [...] I applied the fix to several holes about 5 h ago to writing this review and its already setting up nicely. I'm going to clear the excess coloring tomorrow and reassemble the seat and install them back in my car. Bottom line $10 and i could patch another 10 holes with moderately professional results. Can't beat it."',
  'category': ['Arts Crafts and Sewing', 'Crafting']
}

```

Figure 6: A sample from the Amazon 5x5 dataset. From Zangari et al., 2024.

the task at hand into a multiclass and hierarchical one. Being the dataset with the lowest performance scores in the study, BGC is a monumental problem that SPL-BERT aims to surmount.

```

<book date='2018-08-18' xml:lang='en'>
  <title>Creatures of the Night (Second Edition)</title>
  <body>Two of literary comics modern masters present a pair of magical and disturbing stories of strange creatures who are not quite what they seem! In The Price, a mysterious feline engages in a nightly conflict with an unseen, vicious foe. The Daughter of Owls recounts an eerie tale of a beautiful orphan girl who was found clutching an owl pellet-and how those who would do her wrong would face bizarre, unforeseen consequences. Neil Gaiman (The Sandman, American Gods) delivers his award-winning magic and mystery, realized in Michael Zulli's lavish paintings, newly re-designed in a beautiful new edition!</body>
  <copyright>(c) Penguin Random House</copyright>
  <metadata>
    <topics>
      <d0>Fiction</d0>
      <d1>Graphic Novels & Manga</d1>
    </topics>
    <author>Neil Gaiman</author>
    <published>Nov 29, 2016 </published>
    <page_num> 48 Pages</page_num>
    <isbn>9781506700250</isbn>
    <url> [...] </url>
  </metadata>
</book>

```

Figure 7: Book sample from BGC. From Zangari et al., 2024.

Web of Science

The Web of Science is a paid-access database containing bibliographic information of peer-reviewed studies around the world. Kowsari et al., 2017, as part of their study on hierarchical deep learning for HTC, have collected the abstracts, domain, and keywords of these papers. Each paper’s scientific domain acts as the top-level label, while the keywords provide information for further sub-classifications. Their dataset, WOS, is rather unbalanced, with one out of seven macro-categories taking up a third of total samples, and similarly for the number of sub-classifications. It is a relatively simple dataset for

LLMs, as it is characterized by a 2-level hierarchy and formal, scientific language.

Overall, in terms of textual content, the three datasets above are mostly free from noise, while also having benefited from rough preprocessing. Regarding the actual annotations, however, BGC required further filtering of duplicate labels, where two or more annotations are present on the same level of the hierarchy. Without intervention, this can alter performance scores by being too lenient in raw accuracy scores and not punishing hierarchical violations adequately. Another BGC-specific problem lies in the order of annotations: labels were initially not organized according to level, which caused difficulties when aligning them with the hierarchy. A simple algorithm was created to sort them according to the layer map.

4.4.5 Model training details

BERT-Naive and BERT-MATCH employed roughly the same setup, similarly to the original setups in *htc-survey*. A 12-layer *bert-base-uncased* base was downloaded from HuggingFace and used locally. For the classification task, a classifier layer has been added on top of the base, which takes input from the three last layers before it. The selected learning rate is $2.0e-5$ on all datasets, with a weight decay value of 0.01 for the AdamW optimizer. The models also use early stopping with a patience of 2 and a delta value of 0.0001. The only change is the validation technique (see Experiment 1 for more information). Both were trained on a NVIDIA GeForce RTX 3090: BERT-Naive on 3 epochs, and BERT-MATCH on 4. Each dataset took these models approximately 1-2 hours to finish training and testing.

Meanwhile, SPL-BERT used a similar setup to BERT-Naive in all aspects, except for some SPL-specific details. The model now consists of 12 layers, plus a linear layer of size [2304] to pass the BERT embeddings onto the SPL gating function, two gating layers of size [256], and a final classifier whose size is the number of valid classes depending on the dataset. The main SPL experiment also includes an overparameterization value (S) of 2. SPL-BERT was trained on a Quadro P6000, which may partially explain the longer training time.

4.4.6 Evaluation metrics

htc-survey metrics

All four main metrics of *htc-survey* have been adopted for a fair comparison between current methods and SPL-BERT. These include accuracy, macro-F1, hierarchical macro-F1 (h-fscore) and average hierarchical cost (AHC).

Accuracy and macro-F1 scores are defined in the same way as their standard usage:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \quad (16)$$

$$F1 = 2 \cdot \frac{\text{Pr} \cdot \text{Re}}{\text{Pr} + \text{Re}}, \quad \text{where } \text{Pr} = \frac{TP}{TP + FP}, \quad \text{Re} = \frac{TP}{TP + FN} \quad (17)$$

While accuracy is the raw ratio of correctly predicted samples, it does not truly reflect the imbalance nature of the true label vector (a multi-hot encoding where positive annotations are few), hence unable to penalize false positives adequately. Therefore, HTC metrics also include the macro-F1 score, which is a more balanced measure that accounts for predominantly negative label elements.

Note that accuracy in *htc-survey*, as imported from scikit-learn, concerns all elements in the multi-hot encoding.

The survey also proposed hierarchical metrics. We shall focus on hierarchical F1 (h-F1) Kiritchenko et al., 2006 and AHC.

$$hPr = \frac{\sum_i |\hat{Y}_{aug} \cap Y_{aug}|}{\sum_i |\hat{Y}_{aug}|} \quad (18)$$

$$hRe = \frac{\sum_i |\hat{Y}_{aug} \cap Y_{aug}|}{\sum_i |Y_{aug}|} \quad (19)$$

For the hierarchical versions of F1, Pr and Re, the models' predictions and their targets include their entire path from the root (ancestors), and mispredictions are defined by how much these paths differ.

In simple terms, $\hat{Y}_{aug}$ and Y_{aug} are the extended predictions and targets, respectively. The augmented h-F1 is derived from hierarchical versions of Pr and Re. This metric set is not without drawbacks (Kosmopoulos et al., 2015), particularly overpenalizing predictions

on deeper levels of the hierarchy.

A more recent metric, AHC (Landrieu and Garnot, 2021) has been proposed to compensate for the drawback above. Similarly to our violation count metrics, it is a step towards calculating hierarchical deviations in greater detail. The distance between prediction and target, instead of being comparative, is now calculated as the shortest path taken to reach the target set - in other words, to reach a common ancestor, not the actual deepest target node.

$$\text{AHC}(Y, \hat{Y}) = \frac{1}{|Y|} \sum_j \hat{y}_j \text{dist}(j, Y) \quad (20)$$

This metric is an attempt to capture the semantic similarity between prediction and target (closer nodes in a hierarchy usually imply similarity). It rewards the model for producing predictions that are semantically close to the target - but only when the whole prediction is consistent. In fact, this quickly becomes a critical weakness when the model has no method of strictly enforcing a valid prediction (for instance, the SOTA as of January 2024), which can lead to parent-child mismatches or other violations that greatly undermine its effectiveness. For example, a parent-child pair on opposite sides of the target branch, despite being disconnected, would still yield the exact same cost were they to be on the same branch. The same concern is applicable when the model outputs duplicate answers on one level.

Another criticism to the existing metrics is how they do not fully reflect the variety of hierarchical inconsistencies: mutual exclusivity (including zero and over-one predictions) and parent-child mismatch. It is from this perspective that we propose another set of metrics, simple but effective for controlling hierarchical validity - the main point of HTC but often neglected.

Violation count metrics

The main target of these metrics are mutual exclusivity (ME) errors. Mutual exclusivity is the type of prediction that violates ***Presumption 1***. Violations pertaining to ME can lead to unreliable or inconclusive answers.

Presumption 1: No valid configuration should contain more or less than one positive node on each level.

We further divide ME violations into two cases: zero predictions, where no nodes are marked positive on a certain level, and over-one predictions, where more than one node is predicted as positive. Since these errors can occur at any level of the hierarchy, we have tracked them by each level independently. This explains why the number of violations can far exceed the actual number of samples (see BGC violation results).

For incomplete hierarchies such as BGC, we also provide a gold standard to roughly compare the number of violations with. Due to time constraints, we could not perform more precise zero-prediction counting by directly counting falsely all-zero levels.

To account for other types of violations, we also propose a count for Total Hierarchical Violations (Total HV) and Other Hierarchical Violations (Other HV). Other HV is defined as:

$$\text{Other HV} = \text{Total HV} - ME \quad (21)$$

Total HV includes all rows deviating from the multi-hot encoding dictionary used for assigning multi-hot true labels. Other HV, though far and few, includes mismatches between parent and child nodes. This will provide valuable insight on how exactly the models ensure semantic integrity by respecting hierarchical relationships.

4.5 Experiments

4.5.1 Research questions

In this study, we seek to answer the following questions:

- Are existing methods inconsistent in terms of violation?

Hierarchical metrics such as AHC do not take into account other types of hierarchical violations such as mutual exclusivity. For this purpose, we propose a new set of violation metrics which reflect better the characteristics of hierarchical violations.

- Does SPL improve over the baseline in terms of consistency and accuracy?

We predict that SPL, as a NeSy method which provides reliability and validity in the logical sense, would make a noticeable difference in performance through the reduction of violation counts.